

Hands-On Gurobi Session — Preparation Instructions

To get the most out of our upcoming **Hands-On Gurobi Session** on **Sunday, September 21, 2025**, please complete the following setup tasks **before** the session.

1. Download the Working Folder

Download the materials (ZIP format) from the link below and extract them to a convenient location on your computer:

 [Knapsack Working Folder](#)

2. Install Python

Make sure Python is installed (we recommend version **3.13.7**). Download it from the official website:

 [Python Downloads](#)

☒ *Choose the version that matches your operating system.*

3. Install Gurobi Optimizer

Download and install the latest version of **Gurobi Optimizer** (currently **v12.0.3**) from:

 [Gurobi Optimizer Download](#)

☒ *Make sure it matches your OS and Python version.*

4. Install the Gurobi Python Interface

After Gurobi is installed, open your terminal or command prompt and run:

```
pip install gurobipy
```

5. Activate Your Gurobi License

You need a valid license to run Gurobi. Choose one of the following:

-  **Academic License** (Unlimited usage)

 [Gurobi Academic License](#)

You must register using your university email and activate your license when you are connected under the university domain.

-  **Learner License** (Limited to 2,000 variables and 2,000 constraints)

 [Gurobi Learner License](#)

You can activate your licence even if you are not connected to the university network.

► You can follow this video tutorial to activate your license:

 [How to Activate Gurobi License](#)

6. Verify Your Installation

To confirm Gurobi is properly installed, open your terminal and run:

```
gurobi
```

7. Run the Test Script

- Unzip the working folder.
- Open the script `test_gurobi.py` in your preferred code editor (e.g., VS Code, PyCharm, Jupyter).
- Run the script.

You should see output similar to the following:

```
Academic license – for non-commercial use only – expires 2026-03-26
...
Optimal solution found
x 1.0
y 0.0
z 1.0
Obj: 3.0
```

8. What We'll Cover in the Session

We'll use Gurobi to solve two classic optimization problems in Python:

-  **Knapsack Problem** → [Knapsack.ipynb](#)
-  **1D Bin Packing Problem** → [Binpacking.ipynb](#)

9. Recommended IDE: Visual Studio Code

We'll use **Visual Studio Code** during the session.

-  [Download Visual Studio Code](#)
-  Add the **Python extension** and **Jupyter extension** to run notebooks and scripts easily.

VS Code is lightweight, user-friendly, and supports multiple programming languages through extensions.

✅ **Please complete all steps before the session begins** so we can focus on hands-on modeling and solving problems using Gurobi.

If you run into any issues, feel free to contact us ahead of time.